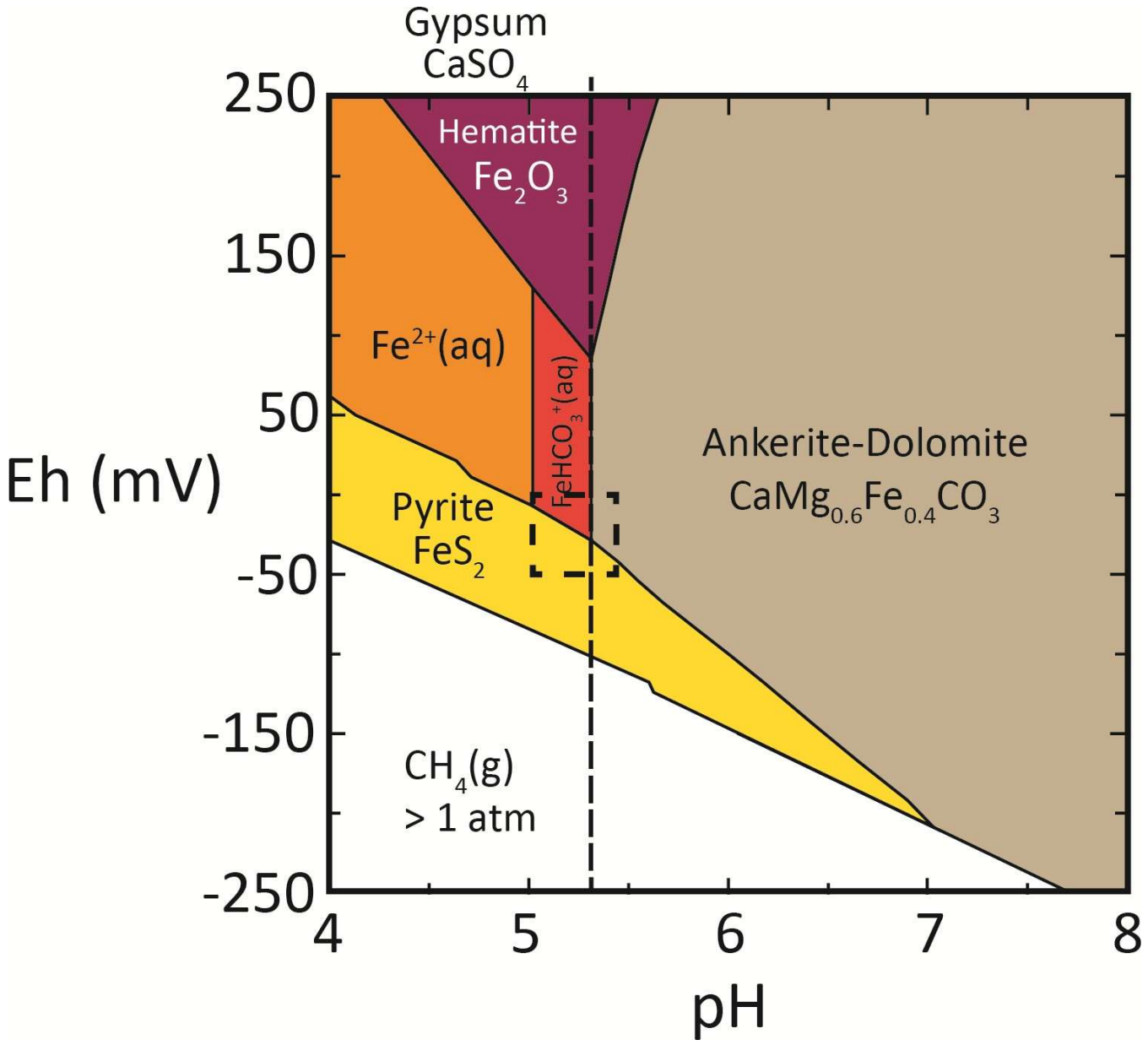
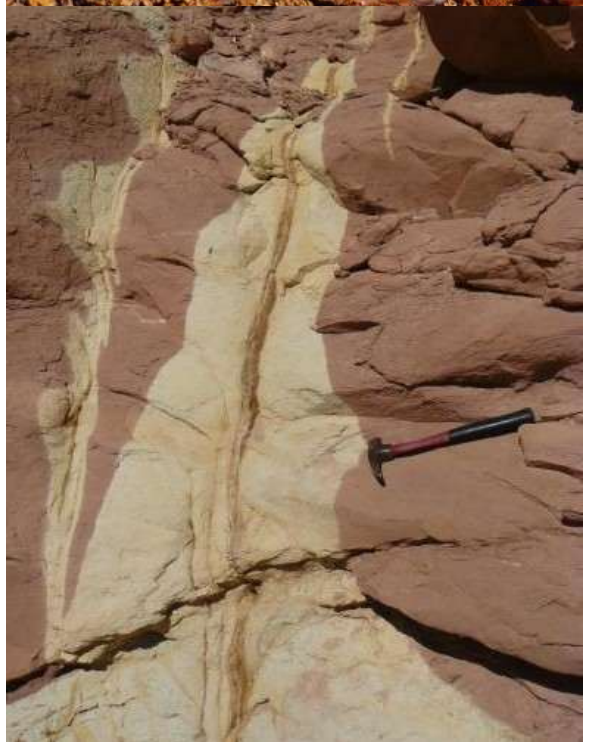
Reduction–oxidation reactions (Redox)







Oxidation-Reduction (Redox) reactions

- Reactions involving a change in oxidation state are redox reactions.
- Oxidation state of an element determines its mobility and toxicity in the environment (e.g As^{3+} reducing conditions and more toxic to humans than As^{5+}).
- Natural waters vary widely in redox conditions; constantly changing an element's chemical speciation and mobility.
- Understanding redox in natural waters (groundwaters) is important for applied geochemistry related to drinking water quality, landfill leachates, organic pollutants, acid mine drainage and radioactive waste containment.
- The major elements which show redox behaviour include C, S, N, and Fe (and Mn in some subsurface systems).
- There are many minor elements which show multiple oxidation states in nature. Some of the important ones include As, Pb, Cu, Cd, Hg, Se, and Cr.

Redox or oxidation state

- Oxidation is electron loss, reduction is electron gain: OIL RIG
- Oxidizing agents accept electrons, and reducing agents give-up electrons.
- The oxidation state (or number) or redox state is the charge that the ions would have if the molecule of species dissociated.
- The oxidation state of an atom in its elemental form is zero (e.g. O_2).
- When bonded to something else, oxygen is in oxidation state 2^- e.g. SO_4^{2-}, CO_3^{2-}, NO_3^- (electron acceptors = oxidising agents)
- Hydrogen is in oxidation state of 1^+ e.g. H_2O, H_2S

Redox systems

- Free electrons do not exist in solution. Any electron lost from one species in solution must be immediately gained by another.

$$Ox_1 + Red_2 \leftrightharpoons Red_1 + Ox_2$$

- Ox_1-Red_1 and Ox_2-Red_2 pairs are known as *redox couples*
- Redox couples are written in the following form:

$$Cu^o\,(s) + Zn^{2+}\,(aq) \leftrightharpoons Cu^{2+}\,(aq) + Zn^o\,(s)$$

- oxidation half-reaction: $Cu^o\,(s) \leftrightharpoons Cu^{2+}(aq) + 2e$
- reduction half-reaction: $Zn^{2+}\,(aq) + 2e \leftrightharpoons Zn^o\,(s)$

Redox reactions

- Consider the redox reaction where: Fe^{2+} is being oxidized to Fe^{3+} (as $Fe(OH)_3$)
  Mn^{4+} (as MnO_2) is being reduced to Mn^{2+}

$$Fe^{2+} \leftrightharpoons Fe^{3+} + e^- \text{ (oxidation) and } Mn^{4+} + 2e^- \leftrightharpoons Mn^{2+} \text{ (reduction)}$$

$$2Fe^{2+} + Mn^{4+} \leftrightharpoons Mn^{2+} + 2Fe^{3+}$$

- Most reactions occur in water, for neutral or acid solutions (H^+)
- The half reactions are written as:

Oxidation $2Fe^{2+} + 6H_2O \leftrightharpoons 2Fe(OH)_3 + 6H^+ + 2e^-$

Reduction $MnO_2 + 4H^+ + 2e^- \leftrightharpoons Mn^{2+} + 2H_2O$

- Combining:

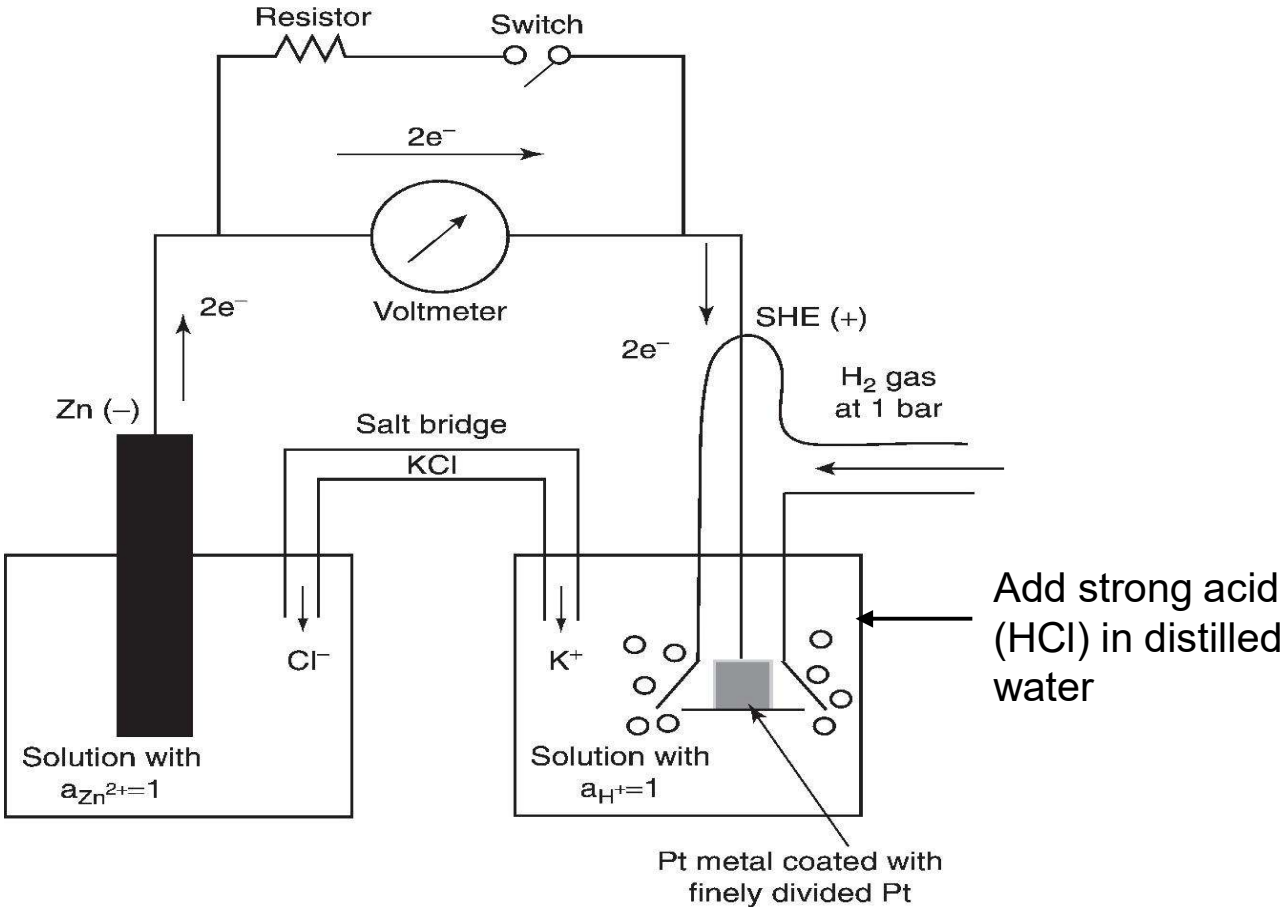
$$2Fe^{2+} + MnO_2 + 6H_2O + 4H^+ + 2e^- \leftrightharpoons 2Fe(OH)_3 + 6H^+ + 2e^- + Mn^{2+} + 2H_2O$$

Gives $\mathbf{2Fe^{2+} + MnO_2 + 4H_2O \leftrightharpoons 2Fe(OH)_3 + 2H^+ + Mn^{2+}}$

- *Balance the atoms apart from oxygen and hydrogen.*
- *Balance the oxygens by adding water molecules.*
- *Balance the hydrogens by adding hydrogen ions.*
- *Balance the charges by adding electrons.*

Electrochemical cell

- If the SHE is connected to another cell containing a different solution, electrons flow is measured as an electromotive force (voltage).



$Zn(s) = Zn^{2+}(aq) + 2e^-$ $2H^+ + 2e^- = H_2\ (g)$

Electromotive series

- Each reaction has the reduced species on the left and oxidized species on the right.
- Half-reactions which cause H^+ to be reduced to H_2 gas have negative values
- Half-reactions which cause H_2 gas to be oxidized to H^+ have positive values
- Strong reducing agents lie at the top of the table and strong oxidizing agents at the bottom.

Reducing agent	Half-reaction	E^0 (volts)
K	$K \Rightarrow K^+ + e^-$	−2.93
Ca	$Ca \Rightarrow Ca^{2+} + 2e^-$	−2.87
Na	$Na \Rightarrow Na^+ + e^-$	−2.71
Mg	$Mg \Rightarrow Mg^{2+} + 2e^-$	−2.36
Mn	$Mn \Rightarrow Mn^{2+} + 2e^-$	−1.18
Zn	$Zn \Rightarrow Zn^{2+} + 2e^-$	−0.76
Fe	$Fe \Rightarrow Fe^{2+} + 2e^-$	−0.47
Ni	$Ni \Rightarrow Ni^{2+} + 2e^-$	−0.24
Pb	$Pb \Rightarrow Pb^{2+} + 2e^-$	−0.13
H_2	$H_2 \Rightarrow 2H^+ + 2e^-$	0.00 (SHE)
Cu^+	$Cu^+ \Rightarrow Cu^{2+} + e^-$	+0.16
Cu	$Cu \Rightarrow Cu^{2+} + 2e^-$	+0.34
Cu	$Cu \Rightarrow Cu^+ + e^-$	+0.52
Fe^{2+}	$Fe^{2+} \Rightarrow Fe^{3+} + e^-$	+0.76
Ag	$Ag \Rightarrow Ag^+ + e^-$	+0.80
Hg	$Hg \Rightarrow Hg^{2+} + 2e^-$	+0.85
Cl^-	$2Cl^- \Rightarrow Cl_2 + 2e^-$	+1.36

*$E^0 = \Delta G_r^0 / n\,F$ (equation 8.22); ΔG_r^0 calculated using $\Delta G^1_{f,\,298.15}$ values in Robie and Hemmingway (1995) listed in Appendix 6.

Table 8.1 An electromotive series listing calculated* standard electrode potentials (E^0) of selected half-reactions arranged in order of decreasing strengths as reducing agents (or of increasing strengths) as oxidizing agents.

Redox potential

- Can calculate the potential for any reaction, e.g. to calculate E^0 for the reaction:

$$Zn + Cu^{2+} \leftrightharpoons Zn^{2+} + Cu$$

- Consider E^0 for each half reaction:

$$Zn = Zn^{2+} + 2e = -0.76 \text{ volt}$$

$$Cu = Cu^{2+} + 2e = +0.34 \text{ volt}$$

- Subtract: -0.76-0.34 = -1.10 volt
- Any reaction with a negative potential will go to the right.
- Any reaction with a positive potential will go to the left.

Reducing agent	Half-reaction	E^0 (volts)
K	$K \Rightarrow K^+ + e^-$	−2.93
Ca	$Ca \Rightarrow Ca^{2+} + 2e^-$	−2.87
Na	$Na \Rightarrow Na^+ + e^-$	−2.71
Mg	$Mg \Rightarrow Mg^{2+} + 2e^-$	−2.36
Mn	$Mn \Rightarrow Mn^{2+} + 2e^-$	−1.18
Zn	$Zn \Rightarrow Zn^{2+} + 2e^-$	−0.76
Fe	$Fe \Rightarrow Fe^{2+} + 2e^-$	−0.47
Ni	$Ni \Rightarrow Ni^{2+} + 2e^-$	−0.24
Pb	$Pb \Rightarrow Pb^{2+} + 2e^-$	−0.13
H_2	$H_2 \Rightarrow 2H^+ + 2e^-$	0.00 (SHE)
Cu^+	$Cu^+ \Rightarrow Cu^{2+} + e^-$	+0.16
Cu	$Cu \Rightarrow Cu^{2+} + 2e^-$	+0.34
Cu	$Cu \Rightarrow Cu^+ + e^-$	+0.52
Fe^{2+}	$Fe^{2+} \Rightarrow Fe^{3+} + e^-$	+0.76
Ag	$Ag \Rightarrow Ag^+ + e^-$	+0.80
Hg	$Hg \Rightarrow Hg^{2+} + 2e^-$	+0.85
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Table 8.1 An electromotive series listing calculated* standard electrode potentials (E^0) of selected half-reactions arranged in order of decreasing strengths as reducing agents (or of increasing strengths) as oxidizing agents.

Eh and pE

- We can measure the potential of a natural solution with respect to the standard hydrogen half-cell, this is the oxidation potential (or redox potential), and is given the symbol **Eh**, usually expressed as **pE**.
- $pE = -\log[e^-]$ (cf $pH = -\log[H^+]$)
- pE is defined as the "activity" of the electron in water. Charge balanced but still has “activity”.
- At equilibrium, an aqueous solution has one pE value and the ratio of oxidized to reduced species activities can be calculated – used to determine speciation and predominance.



Oxidation potential, Eh

- Eh - the potential of a solution relative to the SHE.
- Both pE and Eh measure essentially the same thing. They may be converted via the relationship:

$$pE = \frac{F}{2.303RT} Eh$$

- Where F = 96420 J volt^{-1} eq^{-1} (Faraday's constant - charge of 1 mole of electrons so also in Coulomb's / mole)
- R = 8.3144 JK^{-1} mol^{-1} (Gas constant);
- T = 298K (25°C)

$$pE = 16.9Eh$$

$$Eh = 0.059\,pE$$

- High Eh or high pE of a natural water means the same thing = a high oxidising potential.

Nernst Equation

- For each half-reaction,

  Ox + *n*e$^-$ = Red

- We can define an equilibrium constant

$$K = \frac{[\mathrm{Red}]}{[\mathrm{Ox}][e^-]^n}$$

$$\log[e^-] = \frac{1}{n}\left(\log K - \log\frac{[\mathrm{Red}]}{[Ox]}\right)$$

$$\log[e^-] = \frac{1}{n}\log K - \frac{1}{n}\log\frac{[\mathrm{Red}]}{[Ox]}$$

$$pE = -\log[e^-]$$

$$pE^0 = \frac{1}{n}\log K$$

Rearrange for [e-] gives

$$[e^-]^n = \frac{[\mathrm{Red}]}{K[Ox]} \Rightarrow [e^-] = \sqrt[n]{\frac{\mathrm{Red}}{\mathrm{K[Ox]}}} \Rightarrow [e^-] = \left(\frac{[\mathrm{Red}]}{\mathrm{K[Ox]}}\right)^{\frac{1}{n}} \quad \text{note} \sqrt[n]{\mathrm{x}} = \mathrm{x}^{\frac{1}{n}}$$

$$\log(x^a) = a\log(x)$$

$$pE = pE^o - \frac{1}{n}\log\frac{[\mathrm{Red}]}{[Ox]}$$ the Nernst equation

$$pE = pE^o + \frac{1}{n}\log\frac{[Ox]}{[\mathrm{Red}]}$$

pE, electron Activity

- If pE is **low**, the solution provides a **source** of electrons and is **reducing**.
- If pE is **high**, the solution acts as a **sink** for electrons and is **oxidizing**.
- pE is important for predicting the solubilities of minerals

Example: Mam Tor

- The iron concentration of an aqueous solution, at equilibrium, will be much greater if iron is present in oxidation state 2^+ instead of 3^+.
- Ground water, in contact with iron bearing soils, will dissolve iron as Fe^{2+} (ferrous) ions. When this water, containing Fe^{2+}, contacts air the iron is oxidized to Fe^{3+} (ferric) ions and precipitates at $Fe(OH)_3$.
- In waters with high pE and at a pH of 7, the solubility of Fe^{2+} **fifteen orders** of magnitude greater than Fe^{3+}.
- This is a problem whenever mining operations expose ground water to iron. Ultimately, the ground water will be exposed to an oxidizing environment, which causes the dissolved metals to precipitate. The orange colour of many streams in mining regions is due to the precipitation of ferric hydroxide.

Using pE for predicting speciation

- We want to know the form of copper dissolved in a stream (it could be either Cu^{+} or Cu^{2+}).
- We measure the redox potential of the stream (as an Eh value) as +0.20 V.
- This is more oxidizing than the copper half-reaction $Cu^{+} \leftrightharpoons Cu^{2+} + e^{-}$, for which $E^{0} = +0.16$ V (see table on slide 7).
- Hence we would expect most Cu to be in the form Cu^{2+}.
- The Nernst equation allows us to be quantitative:

$$pE = pE^{o} - \frac{1}{n}\log\frac{[Red]}{[Ox]}$$

$$pE = 16.9Eh = 16.9 \times 0.20 = 3.38$$

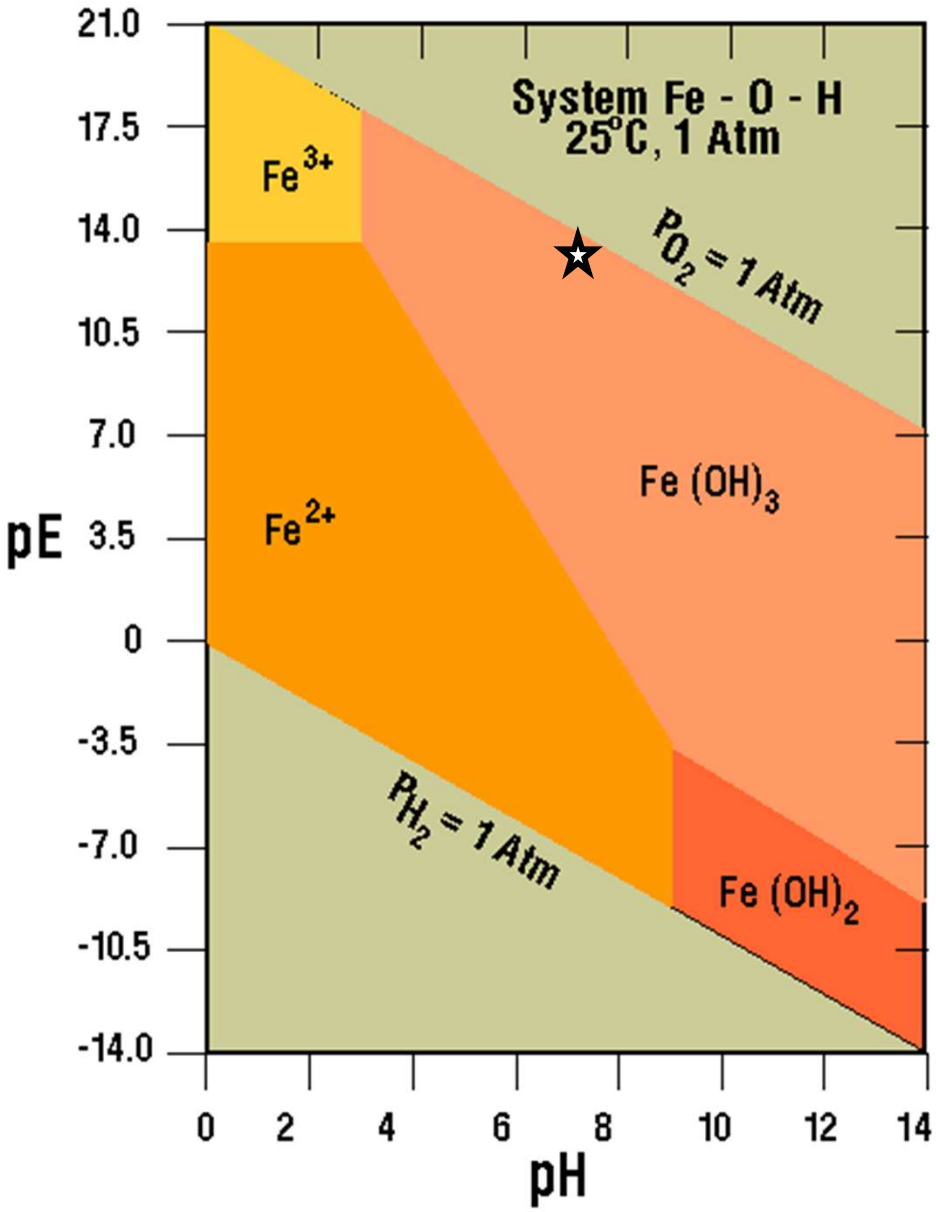
$$pE^{0} = 16.9 \times 0.16 = 2.70$$

$$3.38 = 2.70 - 1 \times \log\frac{[Cu^{+}]}{[Cu^{2+}]}$$

$$3.38 - 2.70 = -\log\frac{[Cu^{+}]}{[Cu^{2+}]}$$

$$\frac{[Cu^{+}]}{[Cu^{2+}]} = \frac{1}{10^{0.68}} = 0.21$$

pE/pH diagrams: Fe-H_2O system



pE/pH diagrams used to predict speciation

¼ O_2 + H^+ + e^- ⇋ ½H_2O pE^o = +20.75

½ H_2 + ⇋ H^+ + e^- pE^o = +0.00

Caution: thermodynamically stable but may be limited by kinetics

H_2: $pE = pE^o - \log([PH_2]^{1/2}/[H^+]) = 0.0 - 0 - pH$

log (A/B) = logA – logB

Natural waters

$pE = 20.75 - \log(0.21)^{¼} - pH$

=13.58 @ pH =7

Predict iron hydroxide stable

Also known as Pourbaix diagrams (Marcel Pourbaix 1904-1998)

pE/pH in natural environments

- pE and pH are both important environmental variables; pE determines oxidation state and both determine species solubility and formation of insoluble oxides (and hydroxides).

- In general, environments in contact with the atmosphere (e.g. rainwater, oceans, streams) have higher pE than environments isolated from the atmosphere (e.g. peat bogs).

- This makes sense if we think about the actual partial pressure of oxygen in the atmosphere ($PO_2 = 0.21$) – this is much nearer to the upper stability limit of water than the lower stability limit.

- In most natural waters redox conditions are controlled by oxygen, sulphur and iron.

Eh and pH –waters natural waters

Oxic waters - waters that contain measurable dissolved oxygen.

Can have low pH (acid generation from oxidation of sulphur) or high pH (high alkalinity)

Reducing waters (anoxic) - waters that contain both dissolved iron and sulfide.

Can have low pH (sulphur converted to sulphuric acid) or high pH (organic reaction produce bicarbonate)

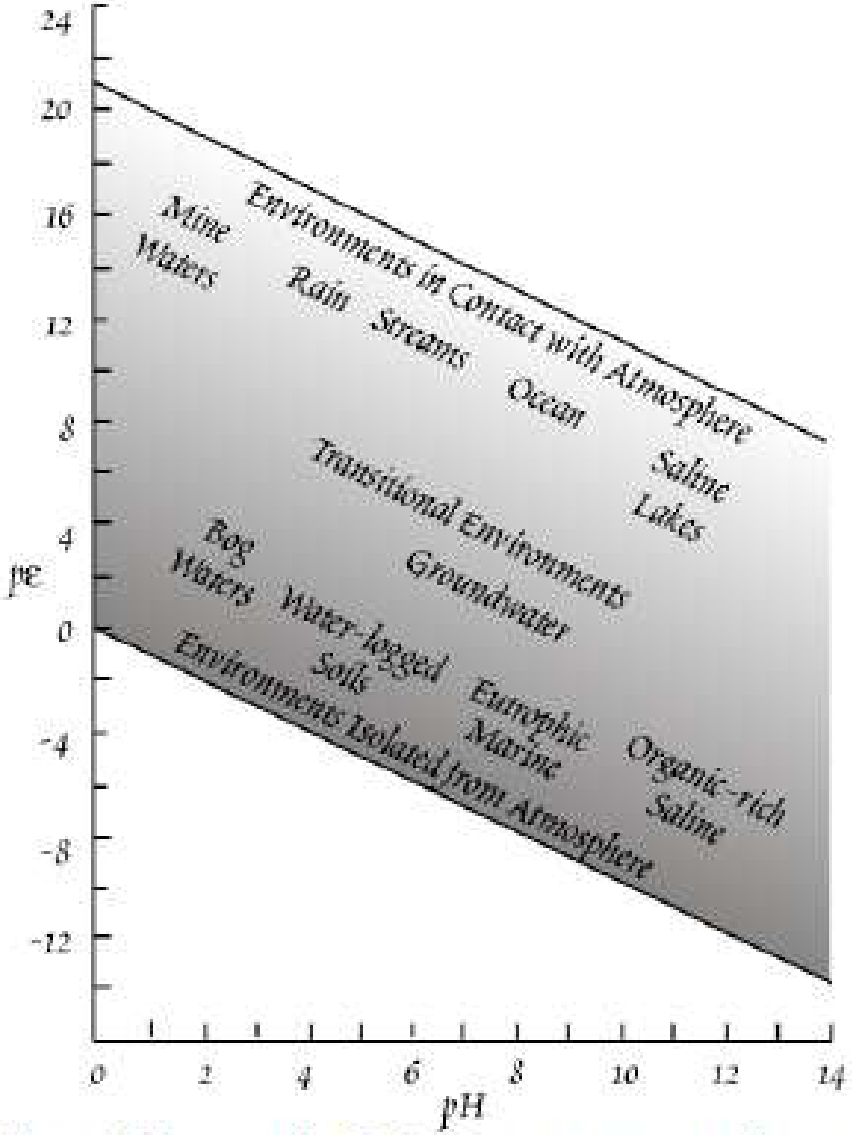


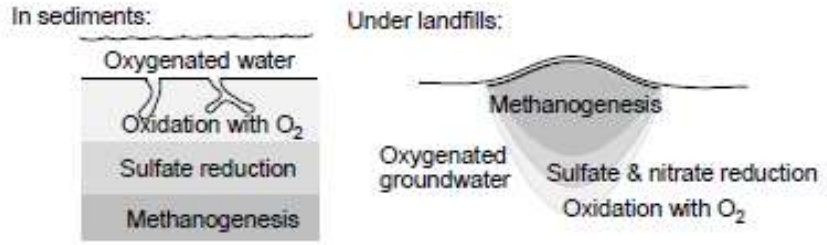
Figure 3.21. pε and pH of various waters on and near the surface of the earth. After Garrels and Christ (1965).

Redox reactions that oxidise organic matter

- Acidic and oxidising at the surface trending to more reducing conditions with depth.
- Oxygen driven at the surface but N>Mn>Fe>S>C with depth (CH_2O represents the simplest form of organic carbon).
- Bacteria mediate most of these reactions and obtain energy for their life processes – becomes increasingly difficult with depth.

	Electron acceptor	Reaction by which organic matter is oxidized	Relative yield of energy	Characteristic H_2 concentration (nM) in solution
1	O^0 in O_2	$CH_2O + O_2 \rightarrow CO_2 + H_2O$	100	0
aerobic conditions				
		Carbonic acid (H_2CO_3)		
anaerobic conditions				
2	N^{5+} in NO_3^-	$5CH_2O + 4NO_3^- \rightarrow 4HCO_3^- + 2N_2 + CO_2 + 3H_2O$	93	<0.1
3	N^{5+} in NO_3^-	$2CH_2O + NO_3^- + H_2O \rightarrow 2HCO_3^- + NH_4^+$		
4	Mn^{4+} in MnO_2	$CH_2O + 3CO_2 + H_2O + 2MnO_2 \rightarrow 2Mn^{2+} + 4HCO_3^-$	87	<0.1
5	Fe^{3+} in $Fe(OH)_3$	$CH_2O + 7CO_2 + 4Fe(OH)_3 \rightarrow 4Fe^{2+} + 8HCO_3^- + 3H_2O$	84	0-0.5
		Fe^{2+} and S^{2-} for formation of pyrite		
6	S^{6+} in SO_4^{2-}	$2CH_2O + SO_4^{2-} \rightarrow H_2S + 2HCO_3^-$	6	1-2
		Reactions that produce bicarbonate (HCO_3^-)		
7	C^0 in CH_2O	$2CH_2O \rightarrow CH_4 + CO_2$ (Methanogenesis)	3	5-10

This series of reactions is relevant in any setting where organic matter is sealed away from the atmosphere:



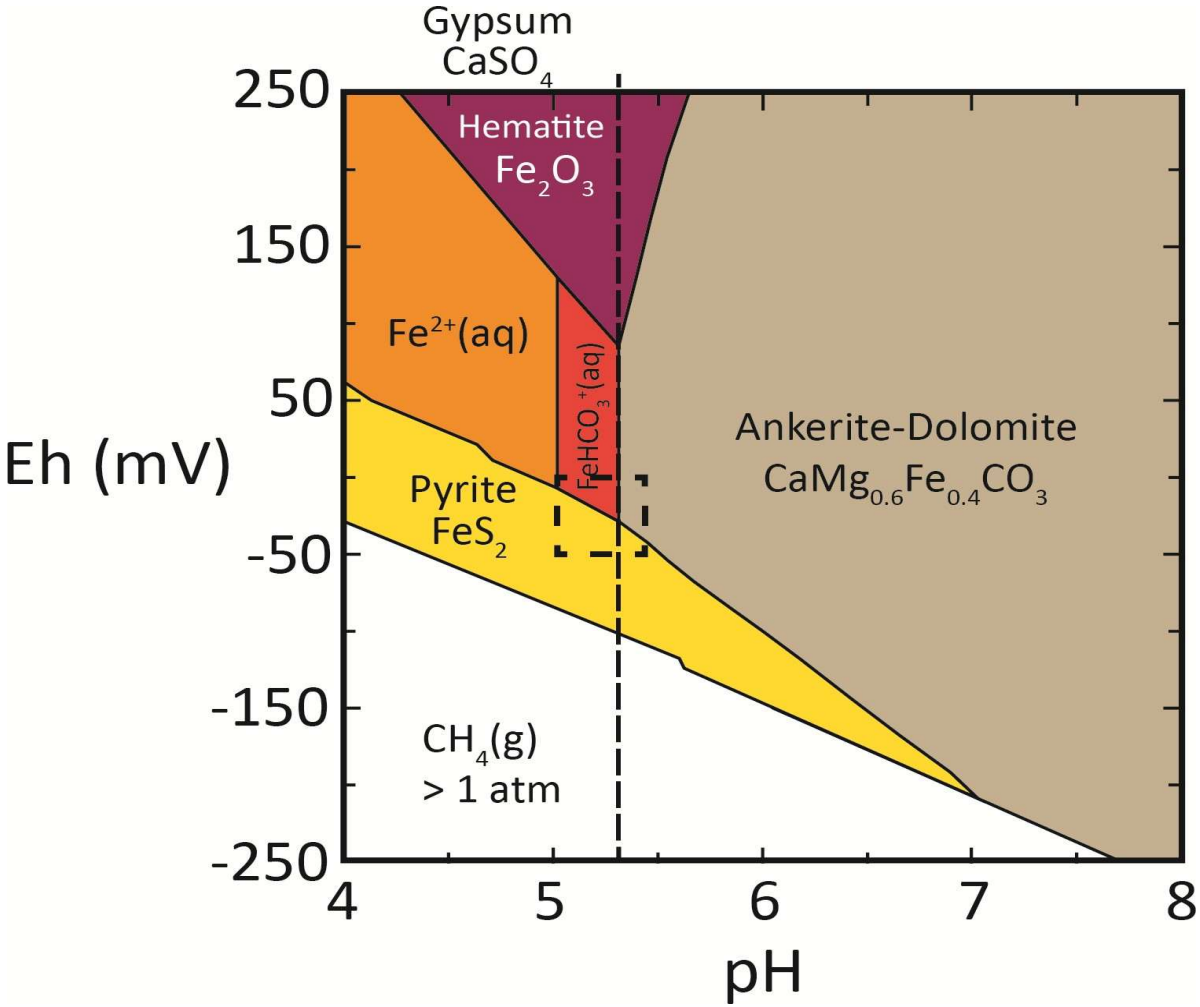
1. Aerobic respiration
2, & 3. Denitrification
4. Manganese reduction
5. Iron reduction
6. Sulphate reduction
7. Methane fermentation

Redox reactions in reservoir rocks





- Reservoir fluids **Na-Cl-SO_4-CO_2 rich**
- Acidic **pH 5.0 to 5.3**
- Reducing **Eh 0 to -50mV at surface**

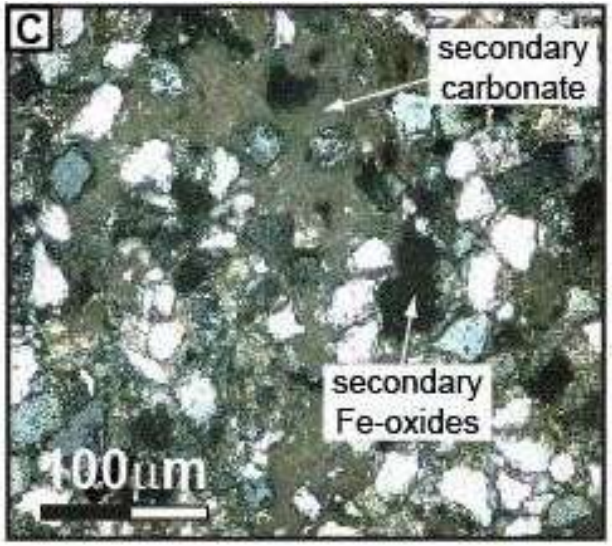


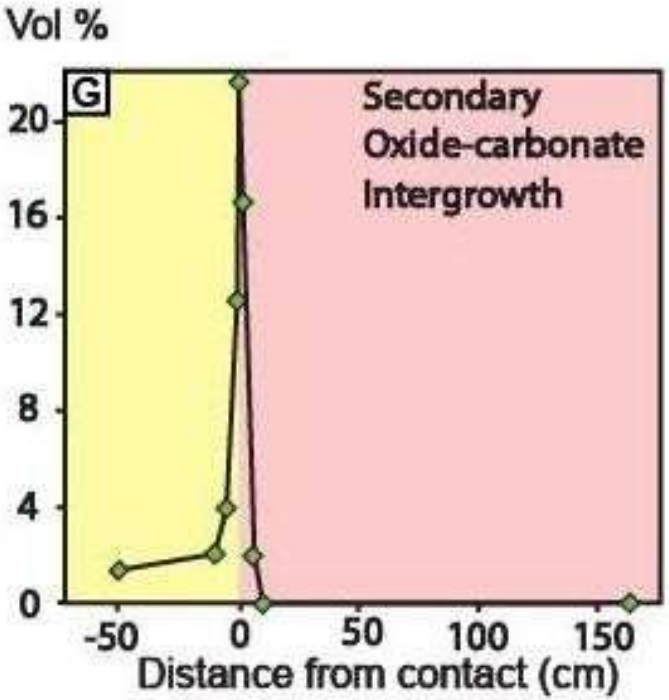
Acidic and reducing fluids dissolve Fe-oxides

Trace metal immobilization at the bleaching reaction front

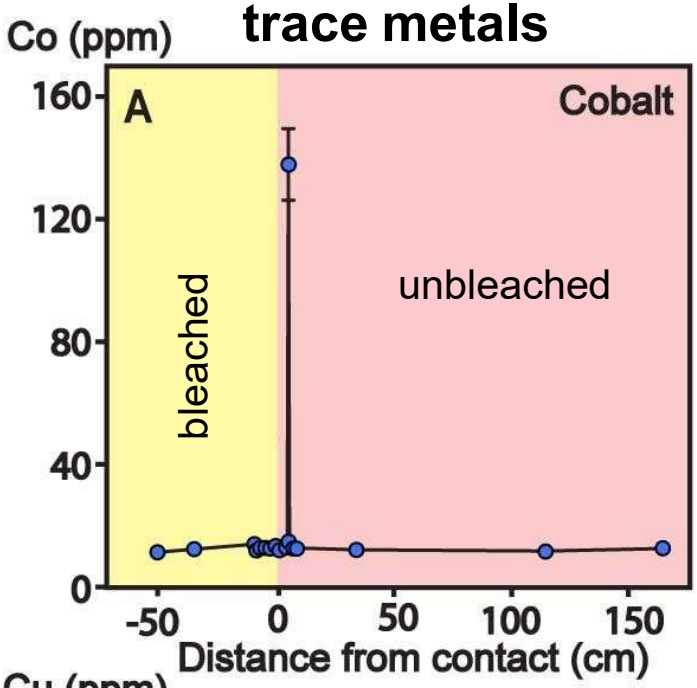
- Carbonates redeposited at reaction front
- Important for reservoir integrity, caprock preservation, harmful metal transport etc..

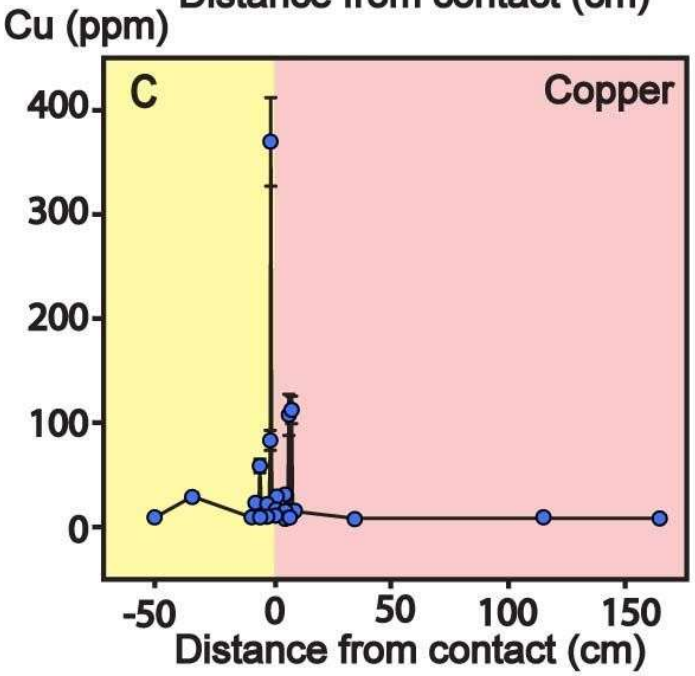
Carbonate Minerals Grow at Front





Co-precipitate trace metals





Summary - 1

- Oxidation is *loss* of electrons, reduction is *gain* of electrons.
- Redox reactions are written in the form of half-reactions for each redox couple.
- Oxidation potential (as measured by Eh or pE) is the redox equivalent of pH. Both Eh and pH are master variables in environmental systems.
- Eh is measured as a voltage (potential or emf) for redox reactions relative to the standard hydrogen electrode (SHE).
- pE is electron activity and is used to represent the redox equilibrium state of natural waters.
- The Nernst equation is used to speciate metals in solution.

Summary - 2

- Plots of Eh (or pE) vs pH are also known as Pourbaix diagrams and are widely used to describe the chemical behaviour species in the hydrosphere, of natural waters and soils.
- Redox is critically important for controlling subsurface conditions that affect everything in the subsurface from deep biosphere to aquifer/reservoir behaviour.

Reading – Misra Chapter 8 especially sections 8.7 and 8.8 on role of microorganisms in oxidation-reduction reactions in the environment